PRODUCT PERFORMACE UPDATE

25%-30% Extended Life of EAF Electrodes at Cleveland Cliffs Mansfield

ELECTRODE DETAILS

- **Electrode Consumption:** Current consumption rates are higher than the industry standard on 24" diameter electrodes. Average heat times are between 2 ½ to 3 hours in length with tap temperatures ranging from 2990° F to 3000° F. Material produced is stainless steel.

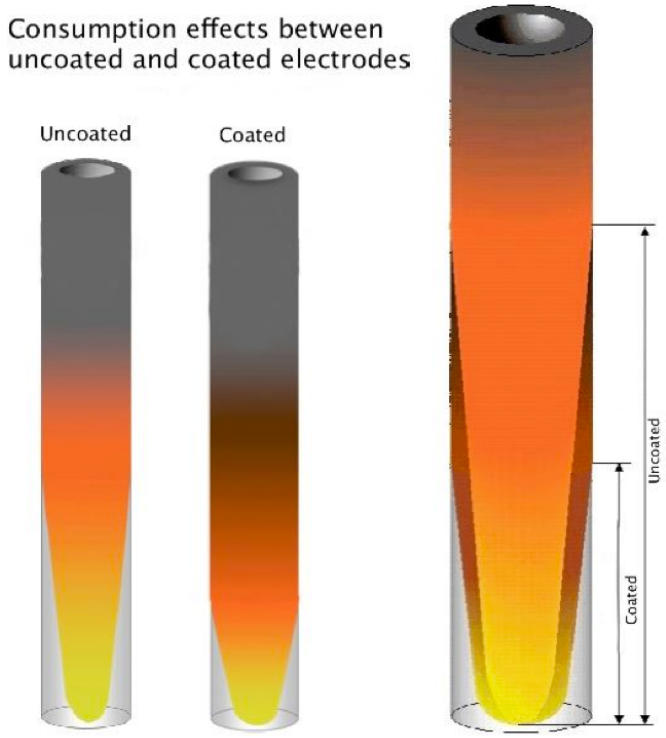
- **Customer Need:** ITC recommended a fully automated spraying solution to reduce electrode consumption rates and lower maintenance/delay time associated with electrode additions.

ITC RECOMMENDATION

- **Engineered Approach:** ITC & EAFab Corporation have entered into a partnership to design and implement a fully automated spraying system to coat electrodes and reduce consumption. ITC recommended a coating specifically design for electrodes, ITC 303 EC, which is formulated for application on carbon/graphite substrates and can be sprayed while the substrate is at elevated temperatures.

- **High Emissivity Coatings:** ITC 303 EC is a proprietary coating designed to work in the most extreme conditions. By using ITC 303 EC, the following benefits were achieved;

  - Reduction of carbon/graphite consumption by 25% to 30%
    - Reduced side oxidation
      1) Reduced oxidation cone
      2) Increased tip diameter
    - Reduced tip consumption
      1) Reduced number of electrode connections
      2) Reduced tip loss
  - Improved conductivity of electrodes
    - By redistribution of the electrical current flow
  - Time savings
    - Improved working conditions for furnace operators and staff
    - Reduced crane time



IMPACT ON THE PROCESS

- REDUCTION IN ELECTRODE CONSUMPTION
  The use of the coating significantly lowered consumption rates by 25% to 30%.

- REDUCED MAINTENANCE & DOWNTIME/DELAY
  With increased electrode life, maintenance/crane time was reduced due to less electrode additions.

- IMPROVED CONDUCTIVITY OF ELECTRODES
  With use of the coating, electrical current flow is now channeled out through the electrode tip and all stray electric arcs are eliminated potentially increasing melt rate and lowering melt times.

- INCREASE IN NUMBER OF HEATS PER MONTH
  Having less delays required for electrode additions and longer electrode life, additional time captured could result in one or several additional heats per month, increasing total yield.